



ADAMAS UNIVERSITY
END SEMESTER EXAMINATION APRIL, 2025
(Academic Session: 2024-25)

Name of the Program:	B.Tech. (Biotechnology/BME/CE/CSE/ECE/ EE/ME)	Semester:	II
Course/Subject Name:	Engineering Mathematics II	Course/Subject Code:	MTH11502
Maximum Marks:	50	Time Duration:	180 Min.
Total No. of Questions:	12	Total No of Pages:	2

1. At top sheet, clearly mention Univ. Roll No., Enrolment No., Course Name & Code, Date of Exam.
2. All parts of a Question should be answered consecutively. Each Answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

Group A

Answer All 5 Questions (5 x 2 = 10)

1	Discuss the convergence of the real sequence $\left\{\frac{(-1)^n}{n} + 1\right\}_{n=1}^{\infty}$.	L1	CO1
2	Find the radius of convergence for the power series $\sum_{n=1}^{\infty} \frac{n!2^n}{n^n} x^n$.	L2	CO2
3	Show that the function $f(z) = \begin{cases} \frac{Re(z)}{ z }, & \text{for } z \neq 0 \\ 0, & \text{for } z = 0 \end{cases}$ is not continuous at $z = 0$.	L3	CO3
4	Check whether the differential equation $(x^2 + y^2 + xy)dx + x(x - 2y)dy = 0$ is exact or not?	L4	CO4
5	If a force $\vec{F}(x, y) = 2x^2y\hat{i} + 3xy\hat{j}$, displaces a particle in the xy -plane from $(0,0)$ to $(1,4)$ along a curve $y = 4x^2$. Find the work done.	L4	CO4

Group B

Answer All 5 Questions (5 x 4 = 20)

6	Find the value of the series $1 + \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \frac{1}{2 \cdot 4} + \frac{1}{4 \cdot 5} + \dots$	L1	CO1
7	Solve the differential equation $\frac{dy}{dx} + \frac{y}{x} = y^2 \ln x$	L2	CO3
8	Find the complete solution of the differential equation $\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 9y = e^{-3x}$.	L3	CO3
9	Using Green's theorem, evaluate $\int_C (x^2y dx + x^2 dy)$, where C is the closed curve described counter clockwise direction of the triangle with the vertices $(0,0)$, $(1,0)$, $(1,1)$.	L2	CO2
10(a)	Show that the function $u(x, y) = 2x + y^3 - 3x^2y$ is a harmonic function. Find the harmonic conjugate of the given function. [2+2]	L6	CO6

OR

10(b)	Find the singular points of the function $f(z) = \frac{\cos(\pi z)}{(z-a)^2}$ and discuss the nature of the singularity. Hence, find the residue of the function. [1+1+2]	L6	CO6
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Group C

Answer All 2 Questions (2 x 10 = 20)

11(a)	i) Show that the function $f(z) = \begin{cases} \frac{x^2y^5(x+iy)}{x^4+y^{10}}, & z \neq 0 \\ 0, & z = 0 \end{cases}$ satisfy the Cauchy-Riemann equations at $z = 0$ but not analytic at $z = 0$. ii) Find Fourier Cosine series representation of the function $f(x) = k, 0 \leq x \leq 5$; where k is a constant. [6+4]	L4	CO4
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OR

(b)	i) Find the Fourier series representation of the function $f(x) = \begin{cases} \pi + x, & \text{for } -\pi < x < 0 \\ 0, & \text{for } 0 < x < \pi \end{cases}$ ii) Evaluate the following contour integral: $\oint \frac{4-3z}{z(z-1)(z-3)} dz, \quad c: z = \frac{5}{2}$ [6+4]	L4	CO4
2(a)	i) Solve the Cauchy-Euler differential equation $x^2 \frac{d^2 y}{dx^2} - 3x \frac{dy}{dx} + 4y = \cos(\ln x).$ ii) Evaluate $\iint_S \vec{A} \cdot \hat{n} ds$, where $\vec{A} = 18xz\hat{i} - 12\hat{j} + 3y\hat{k}$, and S is the part of the plane $2x + 3y + 6z = 12$ including in the first octant. [5+5]	L5	CO5
OR			
2(b)	(i) Apply the method of variation of parameters to solve the differential equation $\frac{d^2 y}{dx^2} + y = \tan x.$ (ii) State Gauss's Divergence theorem of a vector function $\vec{F}(x, y, z)$ taken around a closed surface S. Use the Gauss's Divergence theorem to evaluate $\iint_S \vec{F} \cdot \hat{n} ds$, where $\vec{F} = 4xz\hat{i} - y^2\hat{j} + yz\hat{k}$, and S is the surface of the cube bounded by $x = 0, x = 1, y = 0, y = 1, z = 0, z = 1$. [5+5]	L5	CO5

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